

Applied Linear Algebra Final Exam 2026

Name _____ Student # _____ (Write clearly!)

Each question is worth 5 points. This is important on the questions that say *More details, more points*. You should try for five details.

1. The matrix A has SVD

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & -1 \\ 0 & 2 & -1 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & -5 & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 2 & -1 \end{bmatrix}$$

what is the best rank 1 approximation of A ?

Solution

Sorry, this is not an SVD decomposition. The singular values should be positive, and the conjugating matrices should be orthogonal (not only having orthogonal columns, but normal columns).

But most people made the best of it and gave the largest rank one outer product:

$$A_2 = (-5) \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} \begin{bmatrix} 0 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 0 & -5 & -10 \\ 0 & 5 & 10 \\ 0 & -10 & -20 \end{bmatrix}$$

some, used the second largest, which is almost as good.

$$A_3 = (3) \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \begin{bmatrix} 0 & 2 & -1 \end{bmatrix}.$$

2. When would you use a low rank approximation of a matrix. (More details more points.)

Solution

- Compression and efficiency. Approximating a big $m \times n$ matrix by a low rank r approximation allows us to write it as a $m \times r$ matrix times a $r \times n$ matrix. This reduces the space we need to hold it, and speeds up calculations that involve it.
- Principle Components. Often the most important action of a transformation matrix, or typical samples of a data matrix, are the singular vectors associated to the singular values that are captured in most low rank approximation. Thus the approximation keeps the 'most important' information of the matrix.
- Noise Reduction. When we expect the matrix should be low rank, but error has made it high rank, we can effectively remove error by making a low rank approximation— the span of the smaller singular values will be mostly error.

3. Use the following QR -decomposition of A to solve $Ax = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$.

$$A = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & 1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ 0 & 1 \end{bmatrix}.$$

Solution

We first compute

$$b' = Q^T b = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \frac{1}{\sqrt{5}} \begin{bmatrix} 0 \\ 5 \end{bmatrix} = \begin{bmatrix} 0 \\ \sqrt{5} \end{bmatrix},$$

and then solve $Rx = b'$ using back substitution, or by using R^{-1} :

$$x = R^{-1}b' = \frac{1}{3} \begin{bmatrix} 1 & -2 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 0 \\ \sqrt{5} \end{bmatrix} = \boxed{\sqrt{5} \begin{bmatrix} -2/3 \\ 1 \end{bmatrix}}.$$

4. How you would approximately compute the eigenvalue λ_m of smallest absolute value, of a big matrix A ? (More details more points.)

Solution

Using QR -iteration, we get a matrix similar to A converging to diagonal with this eigenvalue λ_m appearing on the last diagonal entry. So I would use this. First we use Gram-Schmid to decompose $A = Q_0 R_0$ where Q is orthogonal and R_0 is upper triangular. Then we compute $A_1 = R_0 Q_0$ and use Gram-Schmid to decompose $A_1 = Q_1 R_1$, and continue in this way finding QR decomp $A_i = Q_i R_i$ of $A_i = R_{i-1} Q_{i-1}$. If convergence is slow, we first use Arnoldi iteration to get an almost upper-triangular matrix similar to A . (We would have to do a full rank Arnoldi decomposition though to keep the small eigenvectors.) Or we could shift A_i to $A_i - Id$ where d is a the last diagonal in A_3 say. This will be an approximation of λ_m so $A_i - Id$ would have smallest eigenvalue $\lambda'_m = \lambda_m - d$ which would be much closer to 0, Iteration for it would be much faster. Finding it we add d back on to get λ_m .

Also I gave most marks if you found the corresponding eigenvector first via the inverse power method, and then got it from that. This is not really good for a big matrix, because you have to compute the inverse, but it was better than many of the answers.

5. Give the precision $t = 5$ binary ($\beta = 2$) floating point representation of the number 5.

Solution

The integer binary representation is 101 and so the floating point binary representation is 1.01000×2^2 . (We weren't really clear on what t was, the first digit is always 1, so does not count, t counts the number of digits after the decimal. But I accepted 1.0100×2^2 and even 1.01×2^2 as dropping final 0s is so ingrained.)

6. Explain what a backwards stable fixed point algorithm $\tilde{f}$ for a problem f is, and then show that the floating point sum $\tilde{f}(a, b) = \text{fl}(a) + \text{fl}(b)$ is a backwards stable algorithm for $f(a, b) = a + b$.

Solution

An algorithm $\tilde{f}$ is backwards stable for $f : X \rightarrow Y$ if for all $x \in X$ there is $\tilde{x} \in X$ with $\frac{\|x - \tilde{x}\|}{\|\tilde{x}\|} = o(\varepsilon_{\text{mach}})$ such that $f(\tilde{x}) = \tilde{f}(x)$.

We show that $\tilde{f}$ is backwards stable. Indeed let $\tilde{a} = \text{fl}(a)$ and $\tilde{b} = \text{fl}(b)$. By the definition of $\varepsilon_{\text{mach}}$ we have that $\frac{\|a - \tilde{a}\|}{\|\tilde{a}\|} < \varepsilon_{\text{mach}}$ and $\frac{\|b - \tilde{b}\|}{\|\tilde{b}\|} < \varepsilon_{\text{mach}}$. Moreover, we have that

$$\tilde{f}(a, b) = \text{fl}(a) + \text{fl}(b) = \tilde{a} + \tilde{b} = f(\tilde{a}, \tilde{b}),$$

showing that $\tilde{f}$ is backwards stable.

(Really though, I meant to say that $\tilde{f}(a, b) = \text{fl}(a) \oplus \text{fl}(b)$. Some people noticed. In this case the proof is longer.)

There are ε_a and ε_b with $|\varepsilon_a|, |\varepsilon_b| < \varepsilon_{\text{mach}}$ such that $\text{fl}(a) = a(1 + \varepsilon_a)$ and $\text{fl}(b) = b(1 + \varepsilon_b)$, and by the FAFPA there is ε with $|\varepsilon| < \varepsilon_{\text{mach}}$ such that $\text{fl}(a) \oplus \text{fl}(b) = (\text{fl}(a) + \text{fl}(b))(1 + \varepsilon)$. So taking $\tilde{a} = a(1 + \varepsilon_a)(1 + \varepsilon)$ and $\tilde{b} = b(1 + \varepsilon_b)(1 + \varepsilon)$, we get

$$\frac{\|a - \tilde{a}\|}{\|\tilde{a}\|}, \frac{\|b - \tilde{b}\|}{\|\tilde{b}\|} < 3\varepsilon_{\text{mach}}.$$

Further, we have

$$\tilde{f}(a, b) = \text{fl}(a) \oplus \text{fl}(b) = (a(1 + \varepsilon_a) + b(1 + \varepsilon_b))(1 + \varepsilon) = \tilde{a} + \tilde{b} = f(\tilde{a}, \tilde{b}),$$

showing that $\tilde{f}$ is backward stable.

7. Where the following is some matrix A to which we have applied the first step of the Householder algorithm, complete the second step to 'upper triangulate' the second column

$$A = \begin{bmatrix} -2 & 1 & 4 \\ 0 & 3 & 2 \\ 0 & 4 & 2 \end{bmatrix}.$$

Solution

We need the householder projection matrix H_a for the column $a = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$.
Normalising $v = a + e_1 \|a\| = (3, 4) + (5, 0) = (8, 4)$ to $u = (2/\sqrt{5}, 1/\sqrt{5})$, this is

$$H_a = I - 2uu^T = \begin{bmatrix} -3/5 & -4/5 \\ -4/5 & 3/5 \end{bmatrix}.$$

Computing

$$\begin{bmatrix} -3/5 & -4/5 \\ -4/5 & 3/5 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ 4 & 2 \end{bmatrix} = \begin{bmatrix} -5 & -14/5 \\ 0 & -2/5 \end{bmatrix}$$

we get

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -3/5 & -4/5 \\ 0 & -4/5 & 3/5 \end{bmatrix} \begin{bmatrix} -2 & 1 & 4 \\ 0 & 3 & 2 \\ 0 & 4 & 2 \end{bmatrix} = \begin{bmatrix} -2 & 1 & 4 \\ 0 & -5 & -14/5 \\ 0 & 0 & -2/5 \end{bmatrix} =: A_2.$$

This last A_2 is the matrix we had to find.